Q1 (4) - Dot and Cross Product

We have, $|\mathbf{a}|=3,|\mathbf{b}|=4$ and $|\mathbf{c}|=5$. It is given that

$a \perp (b+c), b \perp (c+a)$ and $c \perp (a+b)$

$\Rightarrow \quad a\cdot(b+c)=0, b\cdot(c+a)=0$ and $c\cdot(a+b)=0$

$\Rightarrow \quad a\cdot b+a\cdot c=b\cdot c+b\cdot a=c\cdot a+c\cdot b=0$

or $\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{c}+\mathbf{c}\cdot\mathbf{a}=0$ (adding all the above equations)

Now, $\quad |\mathbf{a}+\mathbf{b}+\mathbf{c}|^2=|\mathbf{a}|^2+|\mathbf{b}|^2+|\mathbf{c}|^2+2(\mathbf{a}\cdot\mathbf{b}+\mathbf{b}\cdot\mathbf{c}+\mathbf{c}\cdot\mathbf{a})$

$=3^2+4^2+5^2=50$

$\therefore |\mathbf{a}+\mathbf{b}+\mathbf{c}| \quad =5\sqrt{2}$

Q2 (3) - Dot and Cross Product

Let the required vector be $\mathbf{a}=x\hat{\mathbf{i}}+y\hat{\mathbf{j}}$ It is given that the vector $\mathbf{C}$ is in XY-plane and is perpendicular

$t0\mathbf{b}=4\hat{\mathbf{i}}+3\hat{\mathbf{j}}$

Therefore, $\mathbf{c}=\lambda(3\hat{\mathbf{i}}-\hat{\mathbf{j}})$

$\Rightarrow \quad \hat{c}=\frac{1}{5}(3\hat{i}-4\hat{j})$

Now, $\quad \mathbf{a}\cdot\hat{\mathbf{b}}=1$ and $\mathbf{a}\cdot\hat{\mathbf{c}}=2$

$\Rightarrow \quad (4x+3y=5$ and $3x-4y=10)$

$(4x+3y=5$ and $3x-4y=-10)$

$\Rightarrow \quad (x=2$ and $y=1)\cdot x=-\dfrac{2}{5}$ and $y=\dfrac{11}{5}$

Hence, $\quad a=2\hat{i}-\hat{j}: a=\dfrac{1}{5}(-2\hat{i}+11\hat{j})$

Q3 (2) - Dot and Cross Product

Clearly, AD' is perpendicular to AB and lies in the plane of AB and AD. Therefore,

$\mathrm{AD}'=\mathrm{AB}\times(\mathrm{AB}\times\mathrm{AD})$

$=(\mathrm{AB}\cdot\mathrm{AD})\mathrm{AB}-(\mathrm{AB}\cdot\mathrm{AB})\mathrm{AD}$

$=5(61\hat{\mathrm{i}}-10\hat{\mathrm{j}}-2\hat{\mathbf{k}})$

$\therefore \quad \cos\alpha=\frac{AD\cdot AD'}{|AD||AD'|}$

$=\frac{(-\hat{\mathbf{i}}+2\hat{\mathbf{j}}+2\hat{\mathbf{k}})\cdot(61\hat{\mathbf{i}}-10\hat{\mathbf{j}}-2\hat{\mathbf{k}})5}{\sqrt{(-1)^2+(2)^2+(2)^2}\times5\sqrt{(61)^2+(10)^2+(-2)^2}}$

$=\frac{\sqrt{17}}{9}$

Q4 (2) - Dot and Cross Product

We have,

$\mathrm{BC}+\mathrm{CA}=\mathrm{AB}$

$\Rightarrow \quad \mathbf{a}+\mathbf{b}=-\mathbf{c}$

$\Rightarrow \quad \mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}$

$\Rightarrow \quad a \times(a+b+c)=a \times 0$

$\Rightarrow \quad \mathbf{a} \times \mathbf{b}+\mathbf{a} \times \mathbf{c}=0$

$\Rightarrow \quad \mathbf{a} \times \mathbf{b}=\mathbf{c} \times \mathbf{a}$

Similarly, we can say that

$\mathbf{b} \times \mathbf{c}=\mathbf{c} \times \mathbf{a}$

$\therefore a \times b=b \times c=c \times a$

Q5 (2) - Dot and Cross Product

We have,

$O A=a, O B=10 a+2 b$ and $O C=b$

Since, p denotes the area of quadrilateral $O A B C$.

$\therefore p=\frac{1}{2}|\mathbf{A C} \times \mathbf{O B}|$

$=\frac{1}{2}|(b-\mathbf{a})-(10 \mathbf{a}+2 \mathbf{b})|$

$=\frac{1}{2}|10 \mathbf{b} \times \mathbf{a}-2 \mathbf{a} \times \mathbf{b}|$

$=\frac{1}{2}|10 \mathbf{b} \times \mathbf{a}+2 \mathbf{b} \times \mathbf{a}|=6|\mathbf{b} \times \mathbf{a}|$

Also, q denotes the area of parallelogram with $O A$ and $O C$ as adjacent sides.

$\therefore \quad q=|O C \times O A|=|b \times \mathbf{a}|$

$\therefore \quad p=6 q$

$\Rightarrow \quad k=6$

Q6 (2) - Dot and Cross Product

The equation of two lines are

$(\mathbf{r}-\mathbf{b}) \times \mathbf{a}=0$ and $(\mathbf{r}-\mathbf{a}) \times \mathbf{b}=0$

$\therefore$ $(\mathbf{r}-\mathbf{b})$ is parallel to $\mathbf{a}$ and $(\mathbf{r}-\mathbf{a})$ is parallel to $\mathbf{b}$.

$\therefore \mathbf{r}=\mathbf{b}+\mathbf{p a}$

and $\mathbf{r}=\mathbf{a}+q\mathbf{b}$

For their point of intersection, we have identical values of r. $\therefore \quad p=q=1$ and hence,

$\mathbf{r}=\mathbf{a}+\mathbf{b}=3\hat{\mathbf{i}}+\hat{\mathbf{j}}-\hat{\mathbf{k}}$

Q7 (2) - Dot and Cross Product

The required vector is given by

$\hat{n}=\frac{AB\times AC}{|AB\times AC|}$

Now, $\quad AB\times AC=(b\times a)\times(c-a)$

$=b\times c-b\times a-a\times c+a\times a$

$=b\times c+a\times b+c\times a \quad [\because a\times a=0]$

We also know that,

$$\text{Area of } \Delta ABC=\frac{1}{2}|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}|$$

$$\therefore \quad \hat{n}=\frac{\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}}{|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}|}$$

$$=\frac{\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}}{2\Delta}$$

$$\left[\because \Delta=\frac{1}{2}|\mathbf{a}\times\mathbf{b}+\mathbf{b}\times\mathbf{c}+\mathbf{c}\times\mathbf{a}|\right]$$

Q8 (1) - Scalar Triple Product and Vector Triple Product

$$\text{Volume}=[abc]=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$\therefore \quad \text{Volume}=\begin{vmatrix} 2 & -2 & 0 \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix}$$

$=2(-1+0)+2(-1+3)+0$

$=-2+4=2$

Q9 (4) - Scalar Triple Product and Vector Triple Product

We have, $[a+bb+cc+a]$

$=\begin{vmatrix}1 & 1 & 0\\ 0 & 1 & 1\\ 1 & 0 & 1\end{vmatrix}[\mathbf{abc}]$

$=[1(1-0)-1(0-1)+0(0-1)][\mathbf{abc}]$

$=2[\mathbf{abc}]$

Q10 (2) - Scalar Triple Product and Vector Triple Product

We have, $\Delta=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}^2$

$=[\mathbf{abc}]^2=[(\mathbf{a}\times\mathbf{b})\cdot\mathbf{c}]^2$

$=|\mathbf{a}\times\mathbf{b}|^2|\mathbf{c}|^2=|\mathbf{a}|^2|\mathbf{b}|\sin^2\frac{\pi}{6}|\mathbf{c}|^2$

$=\frac{1}{4}|\mathbf{a}||\mathbf{b}|^2$

$=\frac{1}{4}\left(a_1^2+a_2^2+a_3^2\right)\left(b_1^2+b_2^2+b_3^2\right)$

Q11 (3) - Scalar Triple Product and Vector Triple Product

Given, $\mathbf{a}=\hat{\mathbf{i}}+\hat{\mathbf{j}}+\hat{\mathbf{k}}_i\mathbf{b}=\hat{\mathbf{i}}-\hat{\mathbf{j}}+\hat{\mathbf{k}}$ and $\mathbf{c}=\hat{\mathbf{i}}-\hat{\mathbf{j}}-\hat{\mathbf{k}}$

Now, we have $\quad\mathbf{a}+\lambda\mathbf{b}=(1+\lambda)\hat{\mathbf{i}}+(1-\lambda)\hat{\mathbf{j}}+(1+\lambda)\hat{\mathbf{k}}$

$\therefore$ Projection of $(\mathbf{a}+\lambda\mathbf{b})$ on $\mathbf{c}$

$=\left|\frac{(\mathbf{a}+\lambda\mathbf{b})\cdot\mathbf{c}}{|\mathbf{c}|}\right|=\frac{1}{\sqrt{3}}$ [given]

$\Rightarrow\quad\left|\frac{(1+\lambda)-(1-\lambda)-(1+\lambda)}{\sqrt{1+1+1}}\right|=\frac{1}{\sqrt{3}}$

$\Rightarrow\quad\left|\frac{1+\lambda-1+\lambda-1-\lambda}{\sqrt{3}}\right|=\frac{1}{\sqrt{3}}$

$\Rightarrow\quad\left|\frac{\lambda-1}{\sqrt{3}}\right|=\frac{1}{\sqrt{3}}$

$\Rightarrow\quad\lambda-1=1$

$\Rightarrow\lambda=2$

$\therefore\mathbf{a}+\lambda\mathbf{b}=3\hat{\mathbf{i}}-\hat{\mathbf{j}}+3\hat{\mathbf{k}}$

Q12 (1) - Scalar Triple Product and Vector Triple Product

We have, $\mathbf{a} \cdot \hat{\mathbf{d}} = 0$ and $[\mathbf{b c \hat{d}}] = 0$

$\Rightarrow \mathbf{a} \perp \hat{\mathbf{d}}$ and $\mathbf{b}, \mathbf{c}, \hat{\mathbf{d}}$ are coplanar.

$\Rightarrow \hat{\mathrm{d}} \perp \mathbf{a}$ and $\hat{\mathrm{d}}$ lies in the plane of b and $\mathbf{c}$, we know that the vector $\mathrm{r} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$ is perpendicular to $\mathbf{a}$ and lies in the plane of $\mathbf{b}$ and $\mathbf{c}$.

$\therefore \quad \hat{\mathbf{d}} = \pm \frac{r}{|r|}$

Now, $\quad \mathrm{r} = \mathbf{a} \times (\mathbf{b} \times \mathbf{c})$

$\Rightarrow \quad \mathbf{r} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

$\Rightarrow \quad \mathbf{r} = -(\hat{\mathbf{j}} - \hat{\mathbf{k}}) + (\hat{\mathbf{k}} - \hat{\mathbf{i}}) = -\hat{\mathbf{i}} - \hat{\mathbf{j}} + 2\hat{\mathbf{k}}$

$\therefore \quad \hat{\mathrm{d}} = \pm \frac{\mathrm{r}}{|\mathrm{r}|} = \pm \frac{(-\hat{\mathrm{i}} - \hat{\mathrm{j}} + 2\hat{\mathrm{k}})}{\sqrt{1+1+4}} = \pm \frac{\hat{\mathrm{i}} + \hat{\mathrm{j}} - 2\hat{\mathrm{k}}}{\sqrt{6}}$

Q13 (1) - Scalar Triple Product and Vector Triple Product

We have, $\quad a \times (b \times c) = \frac{b+c}{\sqrt{2}}$

$\Rightarrow \quad (a \cdot c)b - (a \cdot b)c = \frac{1}{\sqrt{2}}b + \frac{1}{\sqrt{2}}c$

$\Rightarrow \quad \left\{\mathbf{a} \cdot c - \frac{1}{\sqrt{2}}\right\} b - \left\{a \cdot b + \frac{1}{\sqrt{2}}\right\} c = 0$

$\Rightarrow \quad o\mathbf{a} + \left\{\mathbf{a} \cdot \mathbf{c} - \frac{1}{\sqrt{2}}\right\} \mathbf{b} + \left\{-(\mathbf{a} \cdot \mathbf{b}) - \frac{1}{\sqrt{2}}\right\} \mathbf{c} = \mathbf{0}$

$\Rightarrow \quad \mathbf{a} \cdot \mathbf{c} - \frac{1}{\sqrt{2}} = 0$

and $\quad -(\mathbf{a} \cdot \mathbf{b}) - \dfrac{1}{\sqrt{2}} = 0 \quad [\because \mathbf{a}, \mathbf{b}, \mathbf{c}$ are non-coplanar $]$

Now, $\quad \mathbf{a} \cdot \mathbf{b} = -\frac{1}{\sqrt{2}}$

$\Rightarrow \quad |\mathbf{a}||\mathbf{b}| \cos\theta = -\frac{1}{\sqrt{2}} \Rightarrow \cos\theta = -\frac{1}{\sqrt{2}}$

$\therefore \quad \theta = \frac{3\pi}{4}$

Q14 (2) - Scalar Triple Product and Vector Triple Product

$|c - a| = 2\sqrt{2}$

$\Rightarrow \quad |c|^2 + |a|^2 - 2a \cdot c = 8$

$\Rightarrow \quad |c|^2 + (\sqrt{9})^2 - 2|c| = 8$

$\Rightarrow \quad |c|^2 - 2|c| + 1 = 0$

$\Rightarrow \quad (|c| - 1)^2 = 0$

$\Rightarrow \quad |c| = 1$

Now, $\quad \mathbf{a} \times \mathbf{b} = 2\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$

$\therefore |a \times b| = \sqrt{4+4+1} = 3$

$\therefore |(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}| = |\mathbf{a} \times \mathbf{b}||\mathbf{c}| \sin 30^{\circ}$

$$3 \times 1 \times \frac{1}{2} = \frac{3}{2}$$

Q15 (1) - Line

Given straight lines $\frac{x-1}{k} = \frac{y-2}{2} = \frac{z-3}{3}$ In vector form, the equation of line

$r_1 = (\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}}) + \lambda(k\hat{\mathbf{i}} + 2\hat{\mathbf{j}} + 3\hat{\mathbf{k}})$

and $\frac{x-2}{3} = \frac{y-3}{k} = \frac{z-1}{2}$

In vector torm, the equation of line

$r_2 = (2\hat{\mathbf{i}} + 3\hat{\mathbf{j}} + \hat{\mathbf{k}}) + \mu(3\hat{\mathbf{i}} + k\hat{\mathbf{j}} + 2\hat{\mathbf{k}})$

Comparing with $r_1 = a_1 + \lambda b_1$ and $r_2 = a_2 + \mu b_2$, we have

$a_1 = \hat{i} + 2\hat{j} + 3\hat{k}, b_1 = k\hat{\mathbf{i}} + 2\hat{j} + 3\hat{k}$

$a_2 = 3\hat{i} + 3\hat{j} + \hat{k}, b_2 = 2\hat{i} + k\hat{j} + 2\hat{k}$

Now, $a_2 - a_1 = \hat{i} + \hat{j} - 2\hat{k}$

$$\therefore \quad b_1 \times b_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ k & 2 & 3 \\ 3 & k & 2 \end{vmatrix}$$

$$= \hat{i}(4-3k) - \hat{j}(2k-9) + \hat{k}\left(k^2-6\right)$$

Since, lines are intersect.

$\therefore$ Shortest distance between these lines is zero, then

$(b_1 \times b_2) \cdot (a_2 - a_1) = 0$

$\Rightarrow \quad (4-3k) - (2k-9) - 2\left(k^2-6\right) = 0$

$\Rightarrow \quad 4 - 3k - 2k + 9 - 2k^2 + 12 = 0$

$\Rightarrow \quad -2k^2 - 5k + 25 = 0$

$\Rightarrow \quad 2k^2 + 5k - 25 = 0$

$\Rightarrow \quad 2k^2 + 10k - 5k - 25 = 0$

$\Rightarrow \quad 2k(k+5) - 5(k+5) = 0$

$\Rightarrow \quad (2k-5)(k+5) = 0$

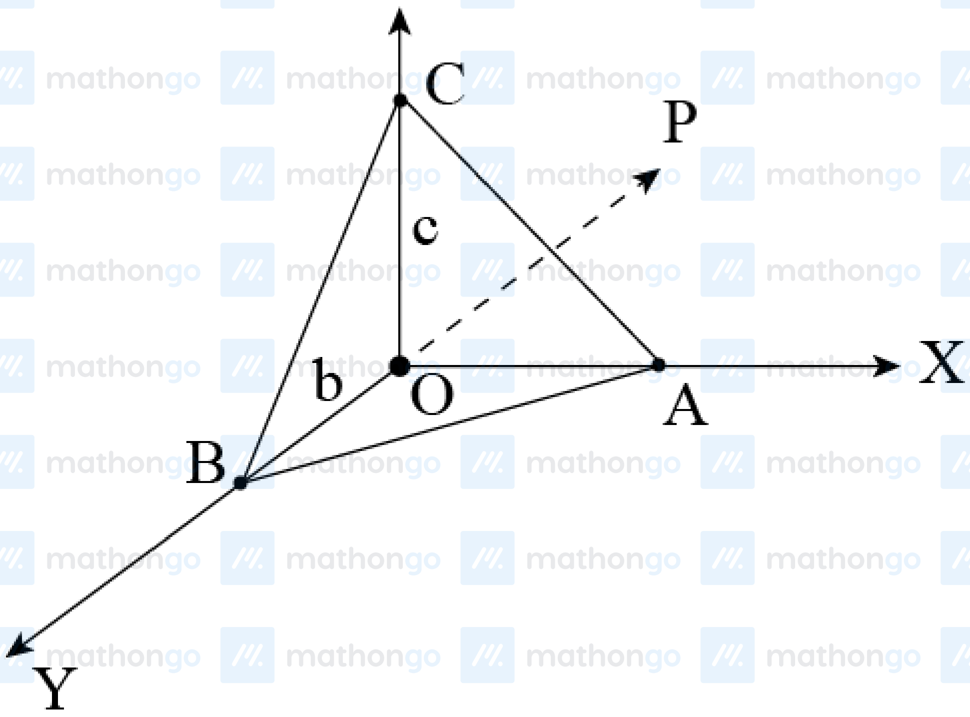
$\Rightarrow \quad 2k - 5 = 0$ or $k + 5 = 0$

$\Rightarrow \quad k = 5/2$ or $k = -5$

Since, k is integer, so $k = -5$

Q16 (1) - Line

Let OP be the normal from O to the plane of $\triangle ABC$ and $l, m_1 n$ be the direction cosines of OP.



Let s be the area of $\triangle ABC$.

$\therefore$ Area of $\Delta OAC =$ Projection of $\triangle HBC$ on ZX -plane $= s \cdot m$ But $\triangle OAC$ is a right angled triangle. Now, the area of $\Delta OAC = \frac{1}{2} ac \Rightarrow \quad \frac{1}{2} ac = s \cdot m$

$\Rightarrow \quad m = \frac{ac}{2s}$

Similarly in $\Delta OBC, \quad l = \frac{bc}{2s}$

$\because \quad l^2 + m^2 + n^2 = 1$

$\left(\frac{ac}{2s}\right)^2 + \left(\frac{bc}{2s}\right)^2 + \left(\frac{ab}{2s}\right)^2 = 1$

$\Rightarrow \quad a^2c^2 + b^2c^2 + a^2b^2 = 4s^2$

$\therefore \quad s = \frac{1}{2}\sqrt{a^2b^2 + b^2c^2 + a^2c^2}$

Q17 (1) - Line

Given, equation of a line in vector form is

$r=(2\hat{i}-\hat{j}+4\hat{k})+\lambda(\hat{i}+\hat{j}-2\hat{k})$

Cartesian form of the given equation is

$\frac{x-2}{1}=\frac{y+1}{1}=\frac{z-4}{-2}$.

Q18 (1) - Line

Equation of line passing through point $A(a,b,c)$ and origin

$$\frac{x-x_1}{x_2-x_1}=\frac{y-y_1}{y_2-y_1}=\frac{z-z_1}{z_2-z_1}$$
$$\Rightarrow \frac{x-0}{a-0}=\frac{y-0}{b-0}=\frac{z-0}{c-0}$$
$$\Rightarrow \frac{x}{a}=\frac{y}{b}=\frac{z}{c}.$$

Since, this line passes through the point $B(a',b',c')$ therefore lies on line.

$\Rightarrow \quad \frac{a'}{a}=\frac{b'}{b}=\frac{c'}{c}$

Q19 (3) - Line

Given lines are

$L_1:\mathbf{r}=(2\hat{\mathbf{i}}+9\hat{\mathbf{j}}+13\hat{\mathbf{k}})+\lambda(\hat{\mathbf{i}}+2\hat{\mathbf{j}}+3\hat{\mathbf{k}})$

and $L_2:\mathbf{r}=(-3\hat{\mathbf{i}}+7\hat{\mathbf{j}}+p\hat{\mathbf{k}})+\mu(-\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}})$

Given lines comparing with

$\mathbf{r}=\mathbf{a}+\lambda\mathbf{b}$

$\mathbf{a}_1=2\hat{\mathbf{i}}+9\hat{\mathbf{j}}+13\hat{\mathbf{k}},\mathbf{b}_1=\hat{\mathbf{i}}+2\hat{\mathbf{j}}+3\hat{\mathbf{k}}$

$\mathbf{a}_2=-3\hat{\mathbf{i}}+7\hat{\mathbf{j}}+p\hat{\mathbf{k}},b_2=-\hat{\mathbf{i}}+2\hat{\mathbf{j}}-3\hat{\mathbf{k}}$

Condition for two given lines to intersects.

$(\mathbf{a}_2-\mathbf{a}_1)\cdot(\mathbf{b}_1\times\mathbf{b}_1)=0$

$$\begin{aligned}\text{Now, }\mathbf{b}_1\times\mathbf{b}_1&=\begin{vmatrix}\hat{\mathbf{i}}&\hat{\mathbf{j}}&\hat{\mathbf{k}}\\1&2&3\\-1&2&-3\end{vmatrix}\\&=\hat{\mathbf{i}}(-6-6)-\hat{\mathbf{j}}(-3+3)+\hat{\mathbf{k}}(2+2)\\&=\hat{\mathbf{i}}(-12)+\hat{\mathbf{k}}(4)\\&=-12\hat{\mathbf{i}}+4\hat{\mathbf{k}}\end{aligned}$$

$$\begin{aligned}\text{and}\quad a_2-a_1&=(-3\hat{\mathbf{i}}+7\hat{\mathbf{j}}+p\hat{\mathbf{k}})-(2\hat{\mathbf{i}}-9\hat{\mathbf{j}}+13\hat{\mathbf{k}})\\&=-5\hat{\mathbf{i}}-2\hat{\mathbf{j}}+(p-13)\hat{\mathbf{k}}\end{aligned}$$

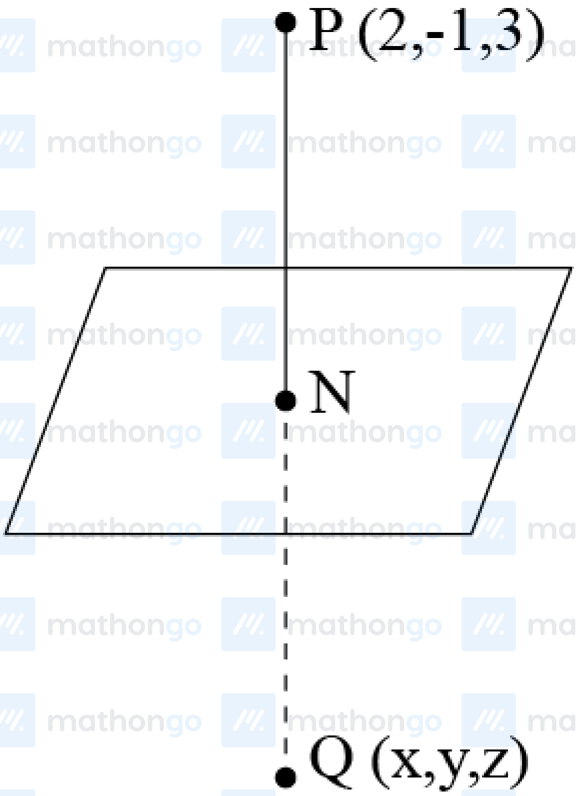
Now, $(\mathbf{a}_2 - \mathbf{a}_1) \cdot (\mathbf{b}_1 \times \mathbf{b}_2) = 0$

$$\Rightarrow \quad [-5\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + (p-13)\hat{\mathbf{k}}] \cdot (-12\hat{\mathbf{i}} + 4\hat{\mathbf{k}}) = 0$$
$$60 + 4(p-13) = 0$$
$$\Rightarrow 60 + 4p - 52 = 0$$
$$\Rightarrow 4p = -8$$
$$\Rightarrow p = -2$$

$\therefore$ Intersecting lines for $p = -2$

Q20 (2) - Plane

Given point $(2, -1, 3)$ and plane $3x - 2y - z = 9$



Let $Q(x, y, z)$ be the image of the point $P(2, -1, 3)$ in the given plane.

The equation of the line through $P(2, -1, 3)$ and perpendicular to given plane

$\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{-1} = r$

The cocrdinates of a general point on this line plane are

$N(3r + 2, -2r - 1, -r + 3)$

$\therefore$ This point lies on the plane, so

$3(3r+2)-2(-2r-1)-(-r+3)=9$

$\Rightarrow \quad 9r+6+4r+2+t-3=9$

$\Rightarrow \quad 14r+6+2-3-9=0$

$\Rightarrow \quad 14r-4=0$

$\therefore \quad r=\frac{2}{7}$

We get, point $N\left(\frac{20}{7},-\frac{11}{7},\frac{19}{7}\right)$

Now, N is the mid-point of PQ.

$\therefore \quad \frac{2+x}{2}=\frac{20}{7}, \frac{-1+y}{2}=\frac{-11}{7}, \frac{3+z}{2}=\frac{19}{7}$

$\Rightarrow 14+7x=40, -7+7y=-22, 21+7z=38$

$\Rightarrow 7x=26 \Rightarrow 7y=-15 \quad \Rightarrow 7z=17$

$\Rightarrow x=\frac{26}{7} \Rightarrow y=-\frac{15}{7} \Rightarrow z=\frac{17}{7}$

Hence, the required image of $P(2,-1,3)$ is

$Q\left(\frac{26}{7},\frac{-15}{7},\frac{17}{7}\right)$

Q21 (4) - Plane

Any point on the line

$$\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}=r$$
$$x=3r+2, y=4r-1, z=12r+2$$

Since, line and plane are intersects So, point lies on the plane $x-y+z=5$.

$(3r+2)-(4r-1)+(12r+2)=5$

$\Rightarrow \quad 3r+2-4r+1+12r+2=5$

$\Rightarrow \quad 11r+5=5$

$\Rightarrow \quad 11r=5-5$

$\Rightarrow \quad 11r=0$

$\therefore \quad r=0$

Intersecting point of fiane and line

$$\begin{aligned}(3r+2,4r-1,12r+2)&=(3\times 0+2,4\times 0-1,12\times 0+2)\\&=(2,-1,2)\end{aligned}$$

So, distance between points $(-1,-5,-10)$ and $(2,-1,2)$.

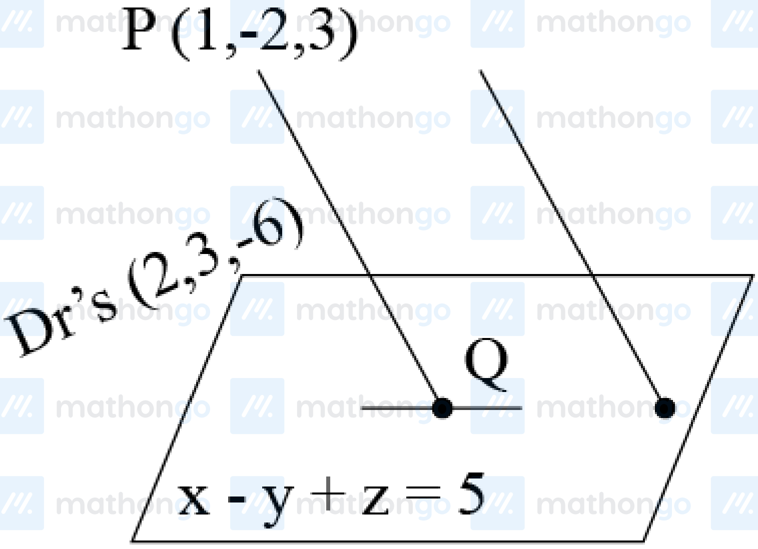
$$\begin{aligned} d &= \sqrt{(2+1)^2+(-1+5)^2+(2+10)^2} \\ &= \sqrt{(3)^2+(4)^2+(12)^2} \\ &= \sqrt{9+16+144} = \sqrt{169} = 13 \end{aligned}$$

Q22 (1) - Plane

$\because$ Equation of PQ, is

$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} = \lambda$

$\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$



$\Rightarrow Q(2\lambda+1, 3\lambda-2, -6\lambda+3)$ lies on $x-y+z=5$

$\Rightarrow 2\lambda+1-3\lambda+2-6\lambda+3=5$

$\therefore \quad \lambda = \frac{1}{7}$ ...(i)

Also, distance $PQ = \sqrt{4\lambda^2+9\lambda^2+36\lambda^2} = 7\lambda = 1$

Q23 (4) - Plane

Let

$\mathbf{b}_1 = 2\hat{\mathbf{i}}+3\hat{\mathbf{j}}-\hat{\mathbf{k}}$
$\mathbf{b}_2 = \hat{\mathbf{i}}-\hat{\mathbf{j}}+2\hat{\mathbf{k}}$

Since, the plane contains the lines which are parallel to the vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ respectively. The normal to the plane $= \mathbf{b}_1 \times \mathbf{b}_2$

$$= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 3 & -1 \\ 1 & -1 & 2 \end{vmatrix}$$

$= \hat{\mathbf{i}}(6-1) - \hat{\mathbf{j}}(4+1) + \hat{\mathbf{k}}(-2-3)$

$= 5\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - 5\hat{\mathbf{k}}$

Acute angle between the vector $2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}}$ and normal to the vector

$\cos\left(\frac{\pi}{2} - \theta\right) = \frac{\mathbf{b}\cdot\mathbf{n}}{|\mathbf{b}||\mathbf{n}|}$

$\sin\theta = \frac{(2\hat{\mathbf{i}} - 2\hat{\mathbf{j}} + \hat{\mathbf{k}})\cdot(5\hat{\mathbf{i}} - 5\hat{\mathbf{j}} - 5\hat{\mathbf{k}})}{\sqrt{(2)^2+(-2)^2+(1)^2}\sqrt{(5)^2+(-5)^2+(-5)^2}}$

$= \frac{10+10-5}{\sqrt{9}\sqrt{75}} = \frac{15}{3\times5\sqrt{3}}$

$\Rightarrow \quad \sin\theta = \frac{1}{\sqrt{3}}$

$\Rightarrow \quad \cot\theta = \sqrt{2}$

$\therefore \quad \theta = \cot^{-1}(\sqrt{2})$

Q24 (2) - Plane

Here, $(1-x)^2 + (1+x)^2 + (1-y)^2 + (1+y)^2 + (1-z)^2 + (z+1)^2 = 10$

$\Rightarrow 2\left[x^2+1\right] + 2\left[y^2+1\right] + 2\left[z^2+1\right] = 10$

$\therefore \quad x^2 + y^2 + z^2 + 3 = 5$

$\Rightarrow \quad x^2 + y^2 + z^2 = 2$

Q25 (1) - Plane

Let the equation of the plane in first system of a rectangular

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

It cuts the coordinate axes at $A(a,0,0), B(0,b,0)$ and $C(0,0,c)$ Let the distance of plane from the origin ρ, then

$p = \left|\frac{\frac{0}{a}+\frac{0}{b}+\frac{0}{c}-1}{\sqrt{\left(\frac{1}{a}\right)^2+\left(\frac{1}{b}\right)^2+\left(\frac{1}{c}\right)^2}}\right|$

$\Rightarrow \quad \rho = \left|\frac{-1}{\sqrt{a^2+\frac{1}{b^2}+\frac{1}{c^2}}}\right|$

$\Rightarrow \quad \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \quad$ (i)

Again, let the equation of plane in second system of rectangular i.e. $\frac{x}{a_1}+\frac{y}{b_1}+\frac{z}{c_1}=1$

It cuts the coordinate axes at $A_1\left(a_1, 0,0\right), B_1\left(0, b_1, 0\right)$ and $C_1\left(0,0, c_1\right)$

Let the distance of the same plane from the origin p, then

$$p=\left|\frac{\frac{0}{a_1}+\frac{0}{b_1}+\frac{0}{c_1}-1}{\sqrt{\left(\frac{1}{a_1}\right)^2+\left(\frac{1}{b_1}\right)^2+\left(\frac{1}{c_1}\right)^2}}\right|$$

$$p=\left|\frac{-1}{\sqrt{\left(\frac{1}{a_1}\right)^2+\left(\frac{1}{b_1}\right)^2+\left(\frac{1}{c_1}\right)^2}}\right|$$

$$=\frac{1}{\sqrt{\left(\frac{1}{a_1}\right)^2+\left(\frac{1}{b_1}\right)^2+\left(\frac{1}{c_1}\right)^2}}$$

$\therefore \frac{1}{p^2}=\frac{1}{\left(a_1\right)^2}+\frac{1}{\left(b_1\right)^2}+\frac{1}{\left(c_1\right)^2}$ ...(ii)

From Eqs. (i) and (ii), we get

$$\frac{1}{a^2}+\frac{1}{b^2}+\frac{1}{c^2}=\frac{1}{\left(a_1\right)^2}+\frac{1}{\left(b_1\right)^2}+\frac{1}{\left(c_1\right)^2}$$

Q26 (4) - Plane

Given lines are concurrent $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}=r_1$ [say] ...(i)

Then, $\quad x=r_1, y=2 q_1, z=3 r_1$

and $\quad \frac{x-1}{3}=\frac{y-2}{-1}=\frac{z-3}{4}=r_2$ [say] ...(ii)

Then, $\quad x=3 r_2+1, y=-r_2+2, z=4 r_2+3$

and $\quad \underline{x+k}=\underline{y-1}=\frac{z-2}{r}=r_3$ [say] ...(iii)

$\frac{x+k}{3}=\frac{y-1}{2}=\frac{z-2}{h}=r_3$

Then, $\quad x=3 r_3-k, y=2 r_3+1, z=h r_3+2$

Given lines are concurrent, so for some values of r_1, r_2 and r_3

Eqs. (i), (ii) and (iii) are same.

From Eqs. (i) and (ii), we get

$\quad r_1=3 r_2+1,2 q_1=-r_2+2$

$\Rightarrow \quad 2\left(3 r_2+1\right)=-r_2+2$

$\Rightarrow \quad 6 r_2+2=-r_2+2$

$\Rightarrow \quad 7 r_2=0 \Rightarrow r_2=0$

Then, $r_1 = 3 \times 0 + 1$

$\Rightarrow \quad r_1 = 1$

So, concurrent point $(x, y, z) = (r_1, 2r_1, 3r_1) = (1, 2, 3)$ Now, from Eqs. (ii) and (iii) which are

$x = 3r_3 - k = 1$

$y = 2r_3 + 1 = 2 \Rightarrow r_3 = \frac{1}{2}$

$z = hr_3 + 2 = 3$

So, $\quad 3 \times \frac{2}{1} - k = 1$

$\Rightarrow \quad k \quad = \frac{3}{2} - 1 \Rightarrow k = \frac{1}{2}$

and $\quad h \times \frac{1}{2} + 2 = 3$

$\Rightarrow \quad \frac{h}{2} = 1 \Rightarrow n = 2$

$\therefore \quad h \quad = 2, k = \frac{1}{2}$

Q27 (2) - Plane

Given equations of the lines are

$r = a_1 + \lambda b$ ...(i)

and $\quad r = a_2 + \mu b$ ...(ii)

The line (i) passes through a point A with position vector $\mathbf{a}_1$ and is parallel to b, the line (ii) passes through a point B with position vector $\mathbf{a}_2$ and parallel to $\mathbf{b}$. $AB =$ (Position vector of B)$-$(Position vector of A)

$= (\mathbf{a}_2 - \mathbf{a}_1)$

$\therefore$ The given lines are coplanar.

$\Rightarrow \mathbf{r}, \mathbf{AB}, \mathbf{b}$ are coplanar

$\Rightarrow \mathbf{r} \cdot (\mathbf{AB} \times \mathbf{b}) = 0$

$\Rightarrow \mathbf{r} \cdot (a_2 - a_1) \times b = 0$

$\Rightarrow [\mathrm{a}_1 \mathrm{a}_2\ \mathrm{b}]$

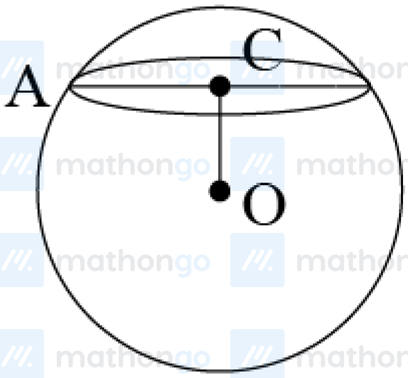
Q28 (4) - Sphere

Given equation of sphere

$x^2 + y^2 + z^2 + 2x - 2y - 4z - 19 = 0$

Comparing with

$x^2+y^2+z^2+2ux+2vy+2wz+d=0$
$2u=2 \Rightarrow u=1$
$2v=-2 \Rightarrow v=-1$
$2w=-4 \Rightarrow w=-2$
$d=-19$



Centre $O(-u,-v_1-w)=O(-1,1,2)$
Radius of sphere $=\sqrt{u^2+v^2+w^2-d}$

$$\begin{aligned}OA&=\sqrt{(1)^2+(-1)^2+(-2)^2-(-19)}\\&=\sqrt{1+1+4+19}=\sqrt{25}=5\end{aligned}$$

$\therefore$ The length of perpendicular from $(-1,1,2)$ to the given plane

$x+2y+2z+7=0$

$$\begin{aligned}OC&=\frac{1\times(-1)+2\times(1)+2\times(2)+7}{\sqrt{(5)^2+(2)^2+2)^2}}\\&=\frac{-1+2+4+7}{\sqrt{9}}=\frac{12}{3}=4\end{aligned}$$

Radius of the circle is given by

$$\begin{aligned}AC&=\sqrt{(OA)^2-OC)^2}=\sqrt{(5)^2-(4)^2}\\&=\sqrt{25-16}=\sqrt{9}=3\end{aligned}$$

Q29 (4) - Sphere

Given sphere

$x^2+y^2+z+4x-2y-6z=155$

Here $2u=4, 2v=-2.2w=-6, d=-155$

$\Rightarrow \quad u=2, v=-1, w=-3$

Centre of sphere $(-u,-v,-w)=(-2,1,3)$

Distance from $(-2,1,3)$ to the plane $12x+4y+3z=327$

$$= \frac{12(-2)+4(1)+3(3)-327}{\sqrt{(12)^2+(4)^2+(3)^2}}$$

$$= \frac{-24+4+9-327}{\sqrt{144+16+9}} \Bigg]$$

$$= \left|\frac{338}{13}\right| = 26$$

$$\therefore \text{ Shortest distance } = 26-\sqrt{4+1+9+155}$$

$$= 26-\sqrt{169}$$

$$= 26-13$$

$$= 13$$

Q30 (3) - Sphere

Required equation of sphere is $\left(x^2+y^2+z^2-25\right)+\lambda x=0$

Passing through $(1,3,-2) \Rightarrow (1+9+4-25)+\lambda \cdot (1)=0$

$\Rightarrow i = 11$

$\therefore$ Equation of sphere is $x^2+y^2+z^2+11x-25=0$